Bridging Arctic Data Gaps Through Co-designed Citizen Science: The ISOSCAN experience

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Introduction

Global warming is rapidly reshaping the Arctic and has significant consequences for the hydrological cycle in the Nordic countries. A warming climate affects precipitation, snow and ice melt, and streamflow in Scandinavia (AMAP, 2021). Heavy snowfall can precondition spring flooding, while rain-on-snow events can induce flash floods. Conversely, low snowfall can lead to summer droughts. Such hydroclimatic extremes endanger ecosystems, infrastructure, and livelihoods. To understand and predict such extremes, and to enable effective adaptation in a warming Arctic, improved hydrological forecasting models are required. In turn, improving these models requires observations of the underlying physical processes. However, knowledge gaps and observational limitations persist, particularly with regard to seasonal snowpack extremes and their hydrological responses in Scandinavia. The remote location of the Arctic regions of Scandinavia makes in-situ observations challenging, resulting in sparse data and considerable uncertainty in hydrological modelling and forecasting. Citizen science can help to fill these data

gaps and supplement traditional remote sensing observations in areas where data is otherwise sparse.

The ISOSCAN project, which stands for “Isotope-aided assessment and forecasting of hydroclimatic extremes in Scandinavia through stakeholder co-design”, is based on this premise. The project aims to improve preparedness for hydrological extremes, such as floods and droughts, in Scandinavia by using stable water isotopes in surface snow to gain insight into the efficiency of storms and water pathways in the ground. The novelty lies in the use of stable water isotopes as tracers. Since stable water isotopes are an inherent property of water in all phases and carry the signatures of key processes, such as evaporation and infiltration, collecting simple snow and water samples can improve our understanding of these processes and support more effective water management. Therefore, in ISOSCAN, measurement data is obtained not only from conventional scientific observations, but also through *citizen scientists* collecting snow samples. To this end, a scalable citizen science framework involving recreational nature users (both locals and visitors) and tourism and leisure stakeholders has been co-designed with scientists and prospective citizen scientists. This has resulted in a highly interdisciplinary effort with a multiplicity of interpretations from various actors, which we present briefly in this short statement.

Citizen science for observations

There are many definitions and approaches to citizen science. Typically, it refers to “the general public engagement in scientific research activities when citizens actively contribute to science either with their intellectual effort or surrounding knowledge or with their tools and resources” (Serrano-Sanz et al., 2014, p. 8). By “citizens”, it is generally understood to mean people who are not professionally trained in the disciplines relevant to the specific project. The extent to which citizens are involved varies, but a common approach is to invite them to collect data for scientific purposes. For instance, they may be requested to photograph biodiversity using a digital camera or smartphone, measure precipitation at home, and share these measurements (Fraisl et al., 2022). The large number of participants and their geographical distribution provide scientists with a notable resource for observing phenomena on a large scale and over time at a significantly lower cost than professional observations (Bonney et al., 2009, p.977). However, to ensure that citizen science has societal value as well as scientific value, it is important to ensure that citizens benefit from participation too.

Three learnings from the ISOSCAN project

During the first 1.5 years of the ISOSCAN project, a citizen science framework for collecting surface snow data in Tromsø, Norway, was developed and tested. This provided valuable insights into the potential of citizen science to address data-sparse regions such as the Arctic. ISOSCAN citizen science has two distinguishing characteristics that set it apart from other citizen science

initiatives reported in the literature. Firstly, physical samples (i.e. snow) must be collected and delivered, rather than photos, videos or numerical measurements being taken and sent digitally (Shirk & Bonney, 2018). Secondly, the participation of citizens in sample collection was *co-designed* with professional scientists and prospective citizen scientists in an iterative process (Figure 1). This approach was adopted to ensure that the resulting citizen science framework would encompass the interests of all stakeholders: the collection of scientifically valuable samples, promoting the motivation of citizen scientists to participate, and broader societal value – scientifically valuable samples, motivation for citizens to engage as well as broader societal value .

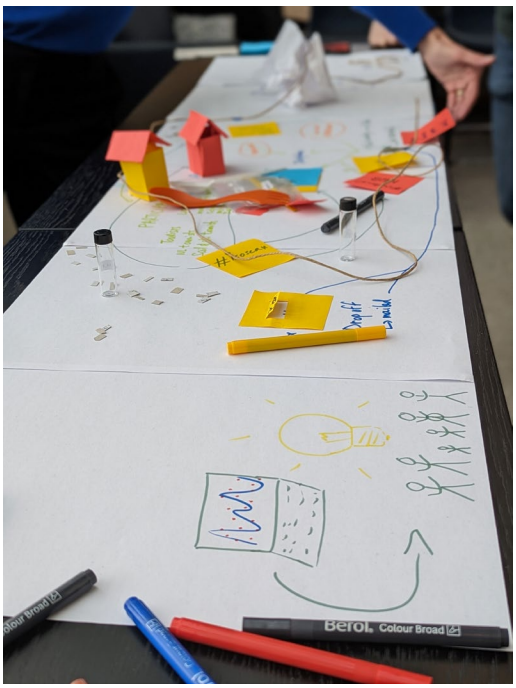


Figure 1: In the first co-design intervention, professional project scientists were invited to map the journey of a stable water isotope using paper, markers, thread, and other physical materials.

One key finding was the recognition of the figurative “distance” between the systematic and structured nature of scientific research, and the experiential, open-ended and highlight-seeking nature of recreational and tourism activities. As the ISOSCAN citizen science concept relies on recreational users in wilderness areas, it must fit into both “worlds” simultaneously. This requires flexibility and compromise on all sides. For example, we have adapted the method of collecting surface snow samples to fit the typical behaviour of a nature user during a day in snow-covered areas. We have also created the “ISOSCAN’er” identity to represent both professional and citizen scientists, and this has been given tangible form in the form of a physical card to address the desire for ownership and 'quick gratification' expressed by prospective citizen scientists (Figure 2).



Figure 2: Citizen science sampling kit for surface snow sampling co-designed in the ISOSCAN project consisting of a yellow zipper closed pouch containing an instruction card, a sampling container, and the ISOSCAN'er card.

Another key finding is the recognition of the multiple groups and interests within a citizen science framework. These groups and interests are more numerous than might be expected. While the interdisciplinarity of project scientists poses sufficient challenges to project progress in itself, there are also several different groups of citizen scientists with their diverse interests and possibilities. For example, local businesses and organisations can play a special role in facilitating and mediating citizen science by leveraging their networks of employees, customers, and members. Although these organisations' owners and employees may not perform the same tasks as those collecting and delivering samples, citizen science projects need to develop ownership and responsibility for these facilitators as well, in order to motivate them to communicate with their own networks.

Finally, setting up and maintaining an effective, large-scale citizen science initiative to support Arctic observation capabilities is not a simple or quick task. It requires resources and expertise. From our experience, working with prospective citizen scientists and facilitators (as discussed above) requires sufficient time to be dedicated to it. It also requires knowledge and experience of working with citizen science itself, beyond the scientific domain in which it may be applied. Therefore, when considering the value of using citizen science for observations, the resources necessary to establish and maintain it should also be taken into account. However, these resources should not only be considered in direct financial terms. Including non-economic value in the early stages of developing citizen science, for example through co-design, can help ensure multidimensional outcomes. The involvement of a large number of citizens can lead to groundbreaking science that would otherwise be impossible. It can also improve societal resilience through better knowledge and preparedness, as well as increasing understanding of science and natural phenomena among non-scientists. It can also improve or even create new local social relations and stewardship for Arctic landscapes. Therefore, in ISOSCAN, the

development of the sampling kit has been approached alongside the long-term involvement of citizen scientists in learning about the water cycle, hydroclimatic extremes, and their forecasts.

Conclusions

We suggest that recreational users, including local residents and tourists, are a valuable yet underutilized resource for generating essential observational data through citizen science initiatives. Based on the experiences of the ISOSCAN project, we have identified three key findings and recommendations regarding the potential of citizen science to address observation gaps in sparsely populated Arctic regions that nevertheless see active use for nature-based recreational activities. These are: recognizing and working to narrow the “distance” between science and everyday recreational contexts; actively involving institutional actors in citizen science processes; and considering multiple dimensions of input resources and the expected value of applying citizen science, including non-economic ones.

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